

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Identifier

Product Description: 5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)

Cat No. : J65100

CAS No 150780-67-9

Molecular Formula C45 H56 N7 O9 P

Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

Details of the supplier of the safety data sheet
Importer

Fisher Scientific Korea
 D5,D6, Incheon Airport Logistics Complex
 150, Gonghangdong-Ro 296 Beon-Gil
 Jung-Gu, Incheon
 Tel: +82-1661-9555
 Fax: +82-2-2023-0603

Supplier

Thermo Fisher Scientific Chemicals, Inc.
 30 Bond Street
 Ward Hill, MA 01835-8099

E-mail address Chem.KR@thermofisher.com

Emergency Telephone Number

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970
CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887
Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the substance or mixture
Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Based on available data, the classification criteria are not met

Environmental hazards

Based on available data, the classification criteria are not met

Label Elements

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5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)

Revision Date 14-Jun-2024

Other Hazards

This product does not contain any known or suspected endocrine disruptors
Toxic to terrestrial vertebrates

NFPA

Health
2

Flammability
1

Instability
0

Physical hazards
-

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	Common Name	CAS No	Index No	Weight %
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	No information available	150780-67-9	Not listed	99 - 100

SECTION 4: FIRST AID MEASURES

Description of first aid measures

General Advice

If symptoms persist, call a physician.

Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur.

Self-Protection of the First Aider

No special precautions required.

Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

Extinguishing media

Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

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Extinguishing media which must not be used for safety reasons

No information available.

Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Nitrogen oxides (NO_x), Hydrogen cyanide (hydrocyanic acid), Oxides of phosphorus.

Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

Environmental precautions

Should not be released into the environment.

Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid contact with skin, eyes or clothing. Avoid dust formation.

Conditions for Safe Storage, Including any Incompatibilities

Store in freezer.

Specific End Uses

Use in laboratories.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

Component	CAS No	Korea	ACGIH TLV	OSHA PEL
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not listed	Not listed	Not listed

Component	CAS No	European Union	The United Kingdom	Germany
5'-O-(4,4'-Dimethoxytrityl)-N	150780-67-9	Not listed	Not listed	Not listed

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5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)

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2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)				
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ACGIH - Biological Exposure Indices

Component	CAS No	ACGIH - Biological Exposure Indices
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not listed

Exposure Controls

Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles

Hand Protection

Protective gloves

Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

Personal protective equipment

Respiratory Protection

Use only those certified by the Korea Occupational Safety and Health Administration.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

Recommended Filter type:

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls

No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance (Physical State, Color, etc.) White - Off-white Solid

Odor

No information available

Odor Threshold

No data available

pH

Not applicable

Melting Point/Range

No data available

Softening Point

No data available

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Boiling Point/Range	No information available	
Flash Point	No information available	Method - No information available

Evaporation Rate	Not applicable	Solid
Flammability (solid,gas)	No information available	
Explosion Limits	No data available	

Vapor Pressure	No data available	
Vapor Density	Not applicable	Solid
Specific Gravity / Density	No data available	
Bulk Density	No data available	
Water Solubility	No information available	
Solubility in other solvents	No information available	

Partition Coefficient (n-octanol/water)

Component	CAS No	log Pow
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	No data available

Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
Viscosity	Not applicable	Solid
Explosive Properties	No information available	
Oxidizing Properties	No information available	

Molecular Formula	C45 H56 N7 O9 P
Molecular Weight	869.94

SECTION 10: STABILITY AND REACTIVITY

Reactivity

None known, based on information available

Chemical Stability

Stable under normal conditions.

Possibility of Hazardous Reactions

Hazardous Polymerization	No information available.
Hazardous Reactions	None under normal processing.

Conditions to Avoid

None known.

Incompatible Materials

None known.

Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO₂). Nitrogen oxides (NO_x). Hydrogen cyanide (hydrocyanic acid). Oxides of phosphorus.

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SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Information on expected route of exposure

Inhalation	Not an expected route of exposure.
Ingestion	No known effect based on information supplied.
Eyes	Not an expected route of exposure.
Skin	No known effect based on information supplied.

Information on Health Hazards

(a) acute toxicity;

Oral	No data available
Dermal	No data available
Inhalation	No data available

Component	CAS No	LD50 Oral	LD50 Dermal	LC50 Inhalation
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	No data available	No data available	No data available

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory	No data available
Skin	No data available

Component	CAS No	Test method	Test species	Study result
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	No data available	No data available	No data available

(e) germ cell mutagenicity; No data available

Component	CAS No	Test method	Test species	Study result
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	No data available	No data available	No data available

(f) carcinogenicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	No data available	No data available	No data available

There are no known carcinogenic chemicals in this product

Component	CAS No	IARC	NTP	ACGIH	OSHA	UK
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-	150780-67-9	Not listed	Not listed	Not listed	Not listed	Not listed

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5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)

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methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)						
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(g) reproductive toxicity; No data available

Component	CAS No	Test method	Test species / Duration	Study result
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	No data available	No data available	No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; Not applicable
Solid

Other Adverse Effects

No information available.

Component	CAS No	EU - Endocrine Disruptors Candidate List	EU - Endocrine Disruptors - Evaluated Substances	Japan - Endocrine Disruptor Information
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

Component	CAS No	Freshwater Fish	Water Flea	Freshwater Algae	Microtox
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	No data available	No data available	No data available	No data available

Persistence and degradability No information available

Bioaccumulative potential No information available

Mobility in soil No information available

Ozone Depletion Potential

Component	CAS No	Ozone Depletion Potential
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not listed

Other adverse effects No information available

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5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)

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SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

SECTION 14: TRANSPORT INFORMATION

Road and Rail Transport

Not Regulated

IATA

Not regulated

IMDG/IMO

Marine Pollutant

Not regulated

No hazards identified

Special Precautions for User

No special precautions required

SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

International Inventories

Component	CAS No	KECL	TSCA	EINECS	IECSC	DSL	NDSL	PICCS	ENCS	ISHL	AICS
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	-	-	-	-	-	-	-	-	-	-

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements	Rotterdam Convention (PIC)	Basel Convention (Hazardous Waste)
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable	Not applicable

Component	CAS No	OECD HPV	Persistent Organic Pollutant	Ozone Depletion Potential
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

Korean National Regulations

Component	CAS No	Act on Registration and Evaluation of Chemical Substances (K-REACH)	Authorised Chemicals	Existing Substances Subject to Registration

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5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable
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Component	CAS No	Chemical Control Act - Toxic Chemicals	Chemical Control Act - Prohibited Chemicals	Chemical Control Act - Use Restricted Chemicals
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

Component	CAS No	Chemical Control Act - Accident Precaution Chemicals (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures)	Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures)
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

Component	CAS No	Waste Control Law	Ministry of Environment - CMR risk	Ministry of Environment - Critically Controlled Substance
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Harmful Agents Subject to Work Environment Monitoring	ISHA - Prohibited substances	ISHA - Substances requiring permission
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Substances subject to control	ISHA - Harmful Agents Requiring Health Examination	ISHA - Permissible Exposure Limits
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

Component	CAS No	ISHA - Subject to Process Safety Reports (minimum quantity)	ISHA - Threshold Limit Values (TLVs) Chemicals	ISHA - Special management materials
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

National Fire Association - Dangerous Substances Minimum quantity requiring a permit

Component	CAS No	Class 1 - Oxidising solids	Class 2 - Flammable solid	Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet	Class 4 - Flammable liquids	Class 5 - Self-reactive substances	Class 6 - Oxidising liquids
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable	Not applicable

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5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)

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Control Parameters

Component	CAS No	Korea	ACGIH - Biological Exposure Indices
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not listed	Not listed

US Management Information

OSHA - Occupational Safety and Health Administration
Not applicable

Component	CAS No	Specifically Regulated Chemicals	Highly Hazardous Chemicals
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable

CERCLA Not applicable

Component	CAS No	CERCLA Extremely Hazardous Substances RQs	Hazardous Substances RQs	SARA 313 - Threshold Values %
5'-O-(4,4'-Dimethoxytrityl)-N2-isobutryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)	150780-67-9	Not applicable	Not applicable	Not applicable

GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Based on available data, the classification criteria are not met.

SECTION 16: OTHER INFORMATION

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

POW - Partition coefficient Octanol:Water

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

Key literature references and sources for data

ALFAAJ65100

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5'-O-(4,4'-Dimethoxytrityl)-N2-isobutyryl-2'-O-methylguanosine-3'-(2-cyanoethyl diisopropylphosphoramidite)

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<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Prepared By	Health, Safety and Environmental Department
Creation Date	15-Apr-2015
Revision Date	14-Jun-2024
Revision Number	6
Revision Summary	New emergency telephone response service provider.

MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet